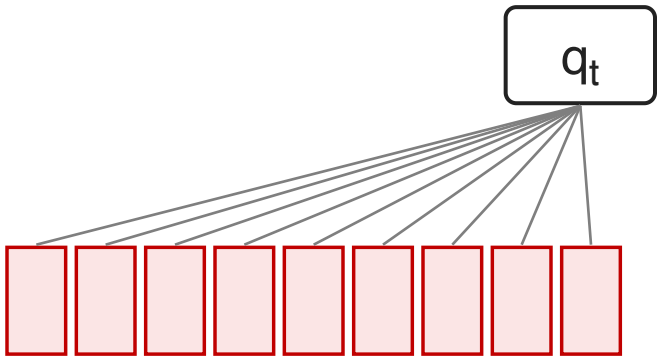
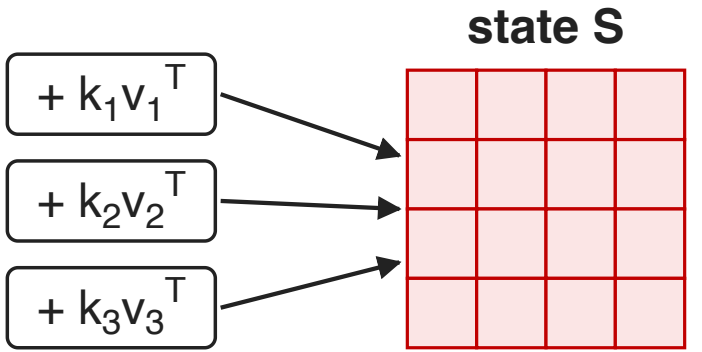
Softmax attention



KV cache: one entry per token

grows with every token

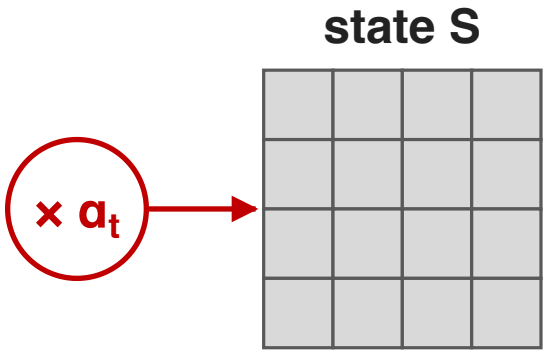
Linear attention



fixed $d_k \times d_v$ matrix

writes pile up without forgetting

Head-wise gate



Mamba2 / GDN: one α per head

same decay for all features